

Specialized Encoders Overfit, A Foundation Encoder Holds: A Cross-Subject Retrieval Degradation Asymmetry in EEG-to-Image Decoding

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Abstract—Inter-subject variability is a central obstacle to generalized decoding in EEG-based Brain-Computer Interfaces (BCIs). The EEG classification literature has repeatedly shown that pretrained foundation encoders generalize across subjects better than task-specific ones. This degradation asymmetry between specialized and foundation encoders has not, to our knowledge, been isolated in the EEG-to-image visual decoding setting. Recent work applies foundation encoders to EEG-to-image decoding and evaluates cross-subject retrieval, but it does so to propose new encoders, calibration, or benchmarks. None compare encoder types with the downstream pipeline held fixed. We fill that gap. Using a 200-way top-1 retrieval metric on THINGS-EEG2, we compare a specialized encoder trained from scratch (ATMS, 3.2M parameters) against a fine-tuned foundation encoder (LaBraM-Base, 6.4M parameters) under matched 10-fold intra-subject and leave-one-subject-out (LOSO) protocols. The specialized encoder achieves higher intra-subject retrieval but collapses cross-subject, retaining only $33.3\% \pm 2.4$ of its performance across three random seeds. The foundation encoder retains $91.5\% \pm 2.7$. The difference in per-subject degradation is highly significant in every seed (paired $t = 9.0$ – 13.7 , all $p < 0.0001$, $n = 10$; Wilcoxon $W = 0$), and the specialized encoder degrades more in all 30 subject-seed pairs. The asymmetry survives a baseline-fair correction: measured per subject as the fraction of *above-chance* performance lost under subject shift, the specialized encoder loses $67.1\% \pm 2.4$ against the foundation encoder’s $2.8\% \pm 3.6$, significant in every seed. On raw LOSO accuracy the two encoders are statistically tied (paired $t = 1.09$, $p = 0.30$). The finding is therefore the degradation asymmetry rather than a raw cross-subject ranking. The absolute LOSO gap between the two encoders is 0.011 (ATMS nominally ahead), which is within noise. We validate that these retrieval numbers reflect genuine visual content through zero-shot image reconstruction.

I. INTRODUCTION

Electroencephalography (EEG) offers accessible access to cognitive states at high temporal resolution, and recent generative models have made it possible to decode perceived images from EEG [12], [2]. A persistent obstacle for deployment is inter-subject variability [13]: a model tuned on one cohort may fail on an unseen individual. Two encoder philosophies address this differently. Specialized encoders

are trained from scratch on the target task and maximize in-domain fidelity; foundation encoders are pretrained on large, heterogeneous EEG corpora and fine-tuned, trading peak in-domain performance for breadth.

In EEG classification, the trade-off is well characterized. In EEG-to-image visual decoding, foundation encoders have been applied [7] and cross-subject generalization has begun to receive attention [8], [9]. The two have not been combined into a controlled comparison. No prior work isolates how a specialized versus a foundation encoder degrades under subject shift with the downstream pipeline held fixed. This paper measures that degradation for both encoder types on a common retrieval metric, and asks which philosophy is more stable under subject shift.

II. RELATED WORK

A. Cross-Subject Generalization of EEG Foundation Models

Foundation encoders such as LaBraM [1] were introduced explicitly to improve cross-subject and cross-dataset transfer relative to task-specific models, and subsequent benchmarks (e.g., AdaBrain-Bench [5] and the EEG Foundation Challenge [6]) prioritize cross-subject transfer and report that foundation models generally outperform specialized baselines on unseen subjects across classification tasks (motor imagery, sleep staging, emotion, auditory attention). The general claim that specialized models overfit to subjects while foundation models generalize is thus established in the classification setting, and we do not claim it as novel.

B. EEG-to-Image Visual Decoding

The visual decoding line, exemplified by NICE [3] and the ATM framework [2], aligns an EEG encoder to CLIP space via a contrastive objective; ATM further reconstructs stimuli through a diffusion prior and image adapter. Recent methods extend this pipeline with multitask learning [10] and hyperbolic feature alignment [11]. Foundation encoders have also been brought directly into this setting: AVDE [7] fine-tunes LaBraM with a contrastive objective for EEG-to-image retrieval and reconstruction, establishing that a pretrained EEG foundation model can serve as the encoder. Cross-subject evaluation has likewise been addressed: SATTC [8] introduces label-free test-time calibration for cross-subject EEG-to-image retrieval under leave-one-subject-out on THINGS-EEG2, and CrossPT-EEG [9] benchmarks cross-participant and cross-time generalization of EEG visual decoding. Our study is distinct from these: we do not propose a new

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encoder, calibration head, or benchmark. Instead, we hold the downstream pipeline fixed and isolate the effect of encoder type: a specialized encoder trained from scratch against a fine-tuned foundation encoder, evaluated on cross-subject retention. We quantify the degradation asymmetry between them with paired statistics. To our knowledge, this comparison of encoder types has not been reported.

III. METHODS

A. Encoders

ATMS (specialized). An Adaptive Thinking Mapper trained from scratch (approximately 3.2M parameters), mapping EEG windows to a 1024-dimensional embedding contrastively aligned to CLIP image space.

LaBraM-Base (foundation). A pretrained EEG foundation model [1] (approximately 6.4M parameters), integrated as a drop-in encoder and fine-tuned end-to-end. Integration used the official channel mapping, resampling from 250 Hz to the model’s native 200 Hz, and a projection from the 200-dimensional native representation to the shared 1024-dimensional CLIP-aligned space.

B. Retrieval Metric and Cross-Subject Protocol

All experiments use THINGS-EEG2 [14], comprising ten subjects with a 200-concept zero-shot test set. We report 200-way top-1 retrieval accuracy (v200-top1): for each test trial, the fraction of cases where the correct concept ranks first among 200 candidates by embedding cosine similarity. We evaluate two regimes for each encoder: intra-subject (train and test on the same subject) and leave-one-subject-out (LOSO, train on nine subjects, test on the tenth), each as a 10-fold sweep across all ten subjects. We define cross-subject retention as the ratio of the ten-fold mean LOSO accuracy to the ten-fold mean intra accuracy, i.e., the fraction of in-domain performance that survives subject shift. We quantify the degradation asymmetry by a paired test (across the ten subjects) of each encoder’s intra-to-LOSO accuracy drop, and, to control for the baseline dependence of the raw drop, we compute a chance-normalized drop per subject and test it with the same paired statistics (Section IV-A).

Training protocol. Both encoders are trained with the AdamW optimizer under an identical schedule: learning rate 2×10^{-4} , batch size 128, a cosine schedule with warmup, and a fixed epoch budget with early stopping. Both encoders are verified to converge *within* this budget rather than being cut off by it: ATMS’s validation loss is flat by approximately epochs 35–40 and LaBraM early-stops near epoch 22. This rules out the alternative explanation that the specialized encoder’s cross-subject collapse is an artifact of undertraining—a control motivated by our own observation that a too-short budget can severely undertrain the slowest encoder and manufacture an apparent cross-subject collapse.

C. Reconstruction Validation

To confirm that retrieval accuracy reflects genuine visual content, we reconstruct images zero-shot from encoder embeddings (subject `sub-08`) via an IP-Adapter over SDXL-

Turbo, conditioning directly on the encoder output (encoder-only, no diffusion prior). We evaluate eight standard reconstruction metrics under a round-based protocol over 200 test concepts. This step is intended only to confirm that the retrieval metric reflects decodable visual content. For compute economy we therefore run it on one subject with the specialized encoder, which is sufficient for that purpose. A broader reconstruction comparison across subjects and encoders is left to future work.

IV. RESULTS

A. Cross-Subject Degradation Asymmetry

Table I and Fig. 1 report the core result. The specialized encoder achieves far higher intra-subject retrieval (10-fold mean 0.323 vs. 0.118) but loses most of it under subject shift, retaining 36% in this seed. The foundation encoder, starting lower, retains 89%. The per-subject intra-to-LOSO drop differs sharply between encoders (mean 0.207 for ATMS vs. 0.014 for LaBraM), and this difference is highly significant under a paired test (paired $t = 11.0$, $p < 0.0001$; Wilcoxon $W = 0$, $p = 0.002$; $n = 10$), with ATMS degrading more than LaBraM in all ten subjects. Table I reports seed 42; the headline figures we quote throughout are the three-seed means of Table II (ATMS $33.3\% \pm 2.4$, LaBraM $91.5\% \pm 2.7$), since those are the values the effect should be judged on.

TABLE I

CROSS-SUBJECT RETRIEVAL DEGRADATION (V200 TOP-1, SEED 42), 10-FOLD MEANS FOR BOTH INTRA-SUBJECT AND LOSO REGIMES. RETENTION IS EXACTLY THE RATIO OF THE TWO TABULATED MEANS. CHANCE IS $1/200 = 0.5\%$, SO LABRAM’S LOSO 0.105 IS ROUGHLY $21 \times$ CHANCE AND ATMS’S 0.116 ROUGHLY $23 \times$.¹

Encoder	Intra	LOSO	Retention	Abs. drop
ATMS (specialized)	0.323	0.116	36%	0.207
LaBraM (foundation)	0.118	0.105	89%	0.014

Two clarifications about how these numbers are formed. First, retention is computed identically in every cell, as the ratio of the ten-fold mean LOSO accuracy to the ten-fold mean intra accuracy, and the parenthetical spreads in Table II are standard deviations *across the three seeds*, not across subjects. We do not average per-subject retention ratios, because near the chance floor that quantity is unstable: the foundation encoder’s intra accuracy is small enough that individual folds can exceed 100% retention and distort the mean. Second, one might object that the absolute intra-to-LOSO drop is baseline-dependent: the specialized encoder’s higher intra accuracy affords a structurally larger drop. We therefore also test the *normalized drop* $D = (\text{intra} - \text{LOSO})/(\text{intra} - c)$ with chance $c = 0.005$, computed per

¹Retention is a raw ratio and is reported for interpretability; because the chance floor (0.005) is small relative to both intra baselines, chance-correcting it changes little. The baseline-fair statistic we *test* is the per-subject normalized drop D below, which is defined against the chance floor explicitly.

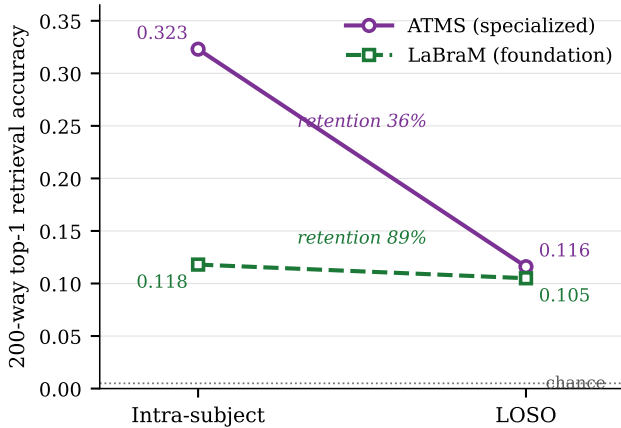


Fig. 1. Intra-subject to leave-one-subject-out degradation (v200 top-1, seed 42, 10-fold means). The specialized encoder starts far higher and collapses; the foundation encoder starts lower and holds. Both land in the same place: the two LOSO endpoints are statistically tied ($t = 1.09$, $p = 0.30$). The dotted line is the 200-way chance floor (0.005).

TABLE II

REPLICATION OF THE DEGRADATION ASYMMETRY ACROSS THREE RANDOM SEEDS (v200 TOP-1). ALL THREE SEEDS SHOW THE SAME EFFECT, WITH ALL TEN SUBJECTS AGREEING IN EVERY SEED, ON BOTH THE RAW DROP AND THE BASELINE-FAIR NORMALIZED DROP D . MEAN (SD) OVER SEEDS IN THE LAST COLUMN.

	Seed 42	Seed 43	Seed 44	Mean (SD)
ATMS retention	36%	33%	31%	33.3% (2.4)
LaBraM retention	89%	92%	94%	91.5% (2.7)
ATMS abs. drop	0.207	0.221	0.240	0.223 (0.017)
LaBraM abs. drop	0.014	0.009	0.007	0.010 (0.004)
Paired t (drop)	11.0	13.67	9.00	11.2 (2.3)
ATMS D	64.4%	67.7%	69.1%	67.1% (2.4)
LaBraM D	6.2%	-0.9%	3.2%	2.8% (3.6)
Paired t (D)	5.16	4.99	7.47	5.9 (1.4)
Subjects ATMS worse	10/10	10/10	10/10	10/10

subject, which divides out the baseline and expresses the fraction of *above-chance* performance lost under subject shift. The asymmetry survives the correction in all three seeds: D is $67.1\% \pm 2.4$ for the specialized encoder against $2.8\% \pm 3.6$ for the foundation encoder, with the paired test significant in every seed ($t = 5.16, 4.99, 7.47$; all $p < 0.001$; Wilcoxon $W = 0$, $p = 0.002$; $n = 10$) and ATMS losing more above-chance performance in all ten subjects of all three seeds. The effect is therefore not an artifact of the encoders’ differing intra baselines. The baseline-fair test is weaker than the test on absolute drops because the foundation encoder’s low intra accuracy leaves a small denominator ($\text{intra} - c$) and so a noisy per-subject D ; we report both rather than the more favourable one.

The foundation encoder’s large per-subject spread in D has a specific and interpretable source: in several subjects its LOSO accuracy exceeds its intra accuracy, giving a negative

D . This is consistent with a structural property of the LOSO protocol, in which the held-out fold is trained on nine subjects and therefore on roughly nine times as much data; for an encoder that is not exploiting subject-specific signal, the additional training data can offset the cost of the subject shift entirely. It is a further reason to read retention alongside the absolute accuracies rather than on its own, and the reason we aggregate retention over folds rather than averaging per-fold ratios.

We stress the correct reading of this table. The finding is the degradation asymmetry: the specialized encoder overfits to subjects while the foundation encoder is stable. It is *not* that the foundation encoder decodes better cross-subject. Its LOSO accuracy (0.105) is in fact slightly below the specialized encoder’s (0.116), and a paired test on raw LOSO accuracy confirms the two are statistically tied ($t = 1.09$, $p = 0.30$, $n = 10$). The claim rests on the retention difference (36% vs. 89%) and the significant gap in per-subject degradation, not on the raw LOSO ranking, which shows parity. Retention of a low intra-subject score is not by itself evidence of strong decoding. The retention figures should be read alongside the absolute accuracies in Table I.

B. Reconstruction Validation

Table III confirms that encoder embeddings carry real visual content: zero-shot encoder-only reconstruction for sub-08 reaches a CLIP two-way identification of 0.80 and AlexNet(5) of 0.84 (chance 0.50), and produces images that preserve coarse visual properties of the stimulus, though not reliably its category. This establishes that the retrieval metric tracks genuine decoding rather than a similarity artifact, while indicating that 63-channel EEG supports coarse visual reconstruction rather than category-faithful synthesis. Arrows denote metric direction ($\uparrow$ higher-better, $\downarrow$ lower-better, distances).

TABLE III

ZERO-SHOT ENCODER-ONLY RECONSTRUCTION, SUB-08 (VALIDATION OF CONTENT CAPTURE).

Metric	Dir.	Value
PixCorr	$\uparrow$	0.1721
SSIM	$\uparrow$	0.3449
AlexNet(2)	$\uparrow$	0.7646
AlexNet(5)	$\uparrow$	0.8440
InceptionV3	$\uparrow$	0.7410
CLIP	$\uparrow$	0.8009
EffNet-B	$\downarrow$	0.8862
SwAV	$\downarrow$	0.5900

C. Cross-Dataset Evaluation: A Data-Availability Boundary

To test whether the degradation asymmetry generalizes beyond THINGS-EEG2, we integrated a second, independent dataset, EEG-ImageNet (later revised as CrossPT-EEG) [9] (62 channels, 1000 Hz, 16 subjects, ImageNet stimuli), whose concepts, subjects, and recording rig all differ from THINGS-EEG2. Because the winter21 ImageNet release

omits a fraction of the referenced stimulus images, and the omissions cluster in specific fine-grained categories, we restricted this analysis to the 70 of 80 categories with at least 75% image availability (3,191 images), preserving category balance. Stimulus embeddings use the same ViT-H-14 model as the main experiments; EEG is cropped to the post-stimulus window, resampled to match the encoder input length, and z-scored per channel to match the input scale of the THINGS-trained encoders. Retrieval is evaluated at the category level (a 70-way maximum), and encoders are compared by cross-subject *retention* rather than raw accuracy, since the two datasets differ in retrieval scale (200-way vs. 70-way).

The integration is complete and the pipeline trains end to end, but held-out retrieval on EEG-ImageNet does not rise above chance. Both encoders fit the training data (training accuracy climbs well above chance; for the foundation encoder it reaches 0.96) while held-out retrieval stays at its floor. We verified this across four configurations: specialized intra-subject, foundation intra-subject, foundation LOSO with roughly fifteen times the training data, and an image-level diagnostic. Consistent with this, the dataset’s own published benchmarks report modest absolute performance (classification accuracies below 0.41 for the strongest baselines).

Three explanations were candidates: the retrieval formulation, a residual methodological fault, or the dataset’s per-trial signal. The image-level diagnostic excludes the first. The second is excluded by the same pipeline producing the strong THINGS-EEG2 results reported above, and independently by a third corpus: we built an equivalent pipeline for Alljoined1 [4], a 64-channel COCO-stimulus EEG dataset with roughly eight repetitions per image, and observed the same signature, training accuracy climbing to 0.97 while held-out retrieval stayed at chance, including when all eight subjects were pooled. We therefore attribute the near-chance result to a weaker per-trial visual signal in these corpora rather than to the method.

This is a boundary worth stating plainly, because it constrains what any study of this kind can currently test. Image-level retrieval requires both repeated presentations of each image and image-level labels. Among the available EEG-image corpora, we find these conditions met only in the THINGS family: other datasets are single-presentation, carry category-level labels only, or (as above) carry too weak a per-trial signal. We consider this documented negative a contribution in its own right: it tells subsequent work which corpora can support image-level retrieval and why the others cannot, and it identifies the concept-space coverage of EEG-image data, rather than encoder design, as the binding constraint on cross-dataset generalization.

V. DISCUSSION

The asymmetry is consistent with the mechanism reported in the EEG classification literature [5]: an encoder trained from scratch with sufficient capacity fits subject-specific idiosyncrasies (electrode placement, individual spectral signatures) that inflate intra-subject accuracy but do not transfer, whereas a foundation encoder pretrained across many

subjects has learned subject-invariant structure and therefore degrades little. Our result shows the same effect surfaces in the visual decoding retrieval setting, where the comparison between the two encoder types had not previously been isolated.

LaBraM’s lower intra-subject peak (0.118 vs. 0.323) fits this account. By not exploiting subject-specific idiosyncrasies, the foundation encoder forgoes the within-subject accuracy that those features provide. The low peak and the high retention are thus two views of the same property. Architectural factors may also contribute. The projection from LaBraM’s 200-dimensional native representation to the 1024-dimensional CLIP space is a narrow bottleneck, and full fine-tuning of a large pretrained model on limited per-subject data may not fully adapt its features to the visual decoding objective. Separating these factors from the subject-invariance effect is left to future work.

The two encoders also differ in capacity (ATMS about 3.2M parameters, LaBraM-Base about 6.4M). This difference does not explain the asymmetry, and it runs against it. Greater capacity would predict more overfitting, yet the larger model is the more stable one. The asymmetry therefore tracks the pretraining regime rather than raw parameter count.

The practical implication is that intra-subject retrieval, the metric usually reported in this line, systematically favors specialized encoders and can mask their poor subject transfer. Retention is the more honest axis when the deployment target is an unseen individual.

VI. LIMITATIONS

The degradation asymmetry replicates across three random seeds (42, 43, 44), with all ten subjects agreeing under every seed (Table II). Reconstruction validation was run for a single subject (sub-08) and for the specialized encoder only; it establishes that the retrieval metric reflects content capture but does not separately validate the foundation encoder’s reconstructions or extend to cross-subject reconstruction. The retention ratio should be read alongside absolute accuracies, since a high retention of a low intra-subject score is not by itself evidence of strong decoding. Finally, the confirmed results are on a single dataset (THINGS-EEG2) with ten subjects, and this is the study’s principal limitation. It is, however, a constraint of the available data rather than of the design: as reported in Section IV-C, image-level retrieval requires repeated presentations and image-level labels, and among the corpora we evaluated only the THINGS family satisfies both. Whether the asymmetry holds on a different concept space therefore remains open, and will remain so until an EEG-image corpus with sufficient per-trial signal outside the THINGS stimulus set becomes available. The comparison also covers two encoders, one specialized and one foundation; establishing the asymmetry as a property of the two model classes rather than of this particular pair requires a broader panel. In follow-on work we have widened the comparison to five encoders spanning a five-fold parameter range across three seeds, and find that retention tracks encoder class rather than size: three specialists spanning 0.64–

3.20M all retain 24–40%, while the two foundation encoders retain more but unequally ($\sim 54\%$ and $\sim 93\%$), indicating that pretraining reliably buys some subject-invariance whose magnitude varies by model. That broader analysis is the subject of a companion report.

VII. CONCLUSIONS

On an EEG-to-image retrieval metric, a specialized encoder retains only $33.3\% \pm 2.4$ of its intra-subject performance under leave-one-subject-out evaluation, while a fine-tuned foundation encoder retains $91.5\% \pm 2.7$ (three random seeds). This degradation asymmetry is established in EEG classification, but to our knowledge it has not previously been isolated as a comparison of encoder types in visual decoding. Here it is significant in every seed (paired $t = 9.0\text{--}13.7$, $p < 0.0001$), with the specialized encoder degrading more in all 30 subject-seed pairs, and it survives a baseline-fair correction that measures the fraction of above-chance performance lost ($67.1\% \pm 2.4$ vs. $2.8\% \pm 3.6$, significant in every seed). Reconstruction confirms the underlying embeddings capture real content. As noted in the Limitations, follow-on work has already widened the comparison to a five-encoder panel; the principal remaining scope gap is therefore generalization to additional *datasets*, which the data-availability boundary (Section IV-C) shows is constrained by the corpus landscape rather than by design. Further future work includes reconstruction validation for the foundation encoder and cross-subject reconstruction.

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